

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2024

Programme: BCSIT

Full Marks: 100

Course: Object Oriented Language (JAVA)

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions. [10×2]

1. Define any two features of OOPs?
2. Define JDK and JVM.
3. Write differences between public, protected, and private access modifiers.
4. Define polymorphism in Java.
5. Define throw and throws.
6. What are the file access modes?
7. What is the difference between AWT and Swing?
8. What is Generic methods?
9. What is static method?
10. Define jump statement.

Section “B”

Descriptive Answer Questions

Attempt **any six** questions. [5×10]

11. Write a program to take 5 number from user and display the min, max and average of the numbers.
12. Write a Java program to create a class Student (name, age, rollNo) and initialize it with data members using a constructor. Display the student record.
13. Define polymorphism and explain the method overloading with example.
14. What is Exception handling in Java? Explain checked exceptions and unchecked exceptions with example code.
15. Differentiate between byte stream and character stream. Write a program to copy the content of a file to another file using streams.
16. What is inheritance? Explain different types of inheritance and also discuss why Java doesn't support multiple inheritance.

Section “C”

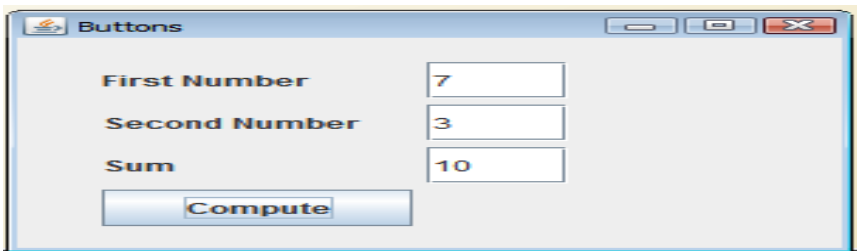
Long Answer Question

Attempt **any two** questions. [2×15]

- 17
 - a. What are Generic in java? How can we write generic class and method?

- b. Write a program using swing to find the sum of two numbers as like in the below image. Frame size:250,120

Frame Title: Calculator



- 18 What is method overriding? How run time polymorphism can be achieved in JAVA? Explain it with suitable example.
- 19 What is layout manager? Discuss different type of layout manger in java.